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Neurobiological degeneracy: Supporting stability, flexibility and pluripotentiality in complex motor skill

John Komar [c,*,1,3], Jia-Yi Chow [b,2], Didier Chollet [a,1], Ludovic Seifert [a,1]

[a] University of Rouen, France
[b] Nanyang Technological University, Singapore
[c] Sheffield Hallam University, United Kingdom

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ABSTRACT

This paper investigated neurobiological degeneracy of the motor system that emerged as a function of levels of environmental constraint. Fourteen participants performed a breaststroke-swimming task that required them to develop a specific biomechanically expert pattern and in turn provide the basis for a suitable task vehicle to study the functional role of movement variability. Inter-limb coordination was defined based on the computation of continuous relative phase between elbow and knee oscillators. Unsupervised cluster analysis on arm–leg coordination revealed the existence of different patterns of coordination when participants achieved the same task goal under different levels of environmental constraints (i.e. different amounts of forward resistances). In addition, clusters differed in terms of higher order derivatives (e.g., joint angular velocity, joint amplitude), suggesting an effective role for degeneracy in learning by allowing the exploration of the key relationships between motor organization and interacting constraints. There is evidence to suggest that neurobiological degeneracy supports the potential for motor re-organization to enhance motor learning.

1. Introduction

From an ecological dynamical perspective (see Seifert, Button, & Davids, 2013 for a review), human beings can be considered as complex dissipative structures with numerous degrees of freedom in a dynamical environment, leading to the emergence of alternative conceptualizations of processes related to perception, cognition, action and decision-making (Williams, Davids, & Williams, 1999). The key emphasis from an ecological dynamical perspective alludes that the emergence of goal-directed behaviour is a consequence of self-organization through the continuous performer–environment interaction (Kelso & Engström, 2006).

Recently, the study of supra-coordinative task (i.e., a task for which the goal is not to produce a specific coordination pattern, Faugloire, Bardy, & Stoffregen, 2009) provides a suitable platform to reconsider the functional role of movement pattern variability (e.g., Hong & Newell, 2006a; Rein, Button, Davids, & Summers, 2010). While movement variability is usually considered as "blameworthy when it comes to human behaviour" (Cohen, Hershberg, & Solomon, 2004, p. 995), there could be a need to consider potential benefits of such variability in motor control and learning (Chen, Liu, Mayer-Kress, & Newell, 2005; Hong & Newell, 2006a; Rein, Davids, & Button, 2010). Indeed, while a linear approach of motor control suggests that movement variability is minimized by taking advantage of the redundancy of the motor system (Harris & Wolpert, 1998), a non-linear approach advocates that degeneracy is more appropriate in examining coordination in neurobiological systems (Newell, Liu, & Mayer-Kress, 2005). In fact, *redundancy* is only one of the multiple ways to promote *degeneracy* in complex neurobiological systems (Mason, 2010). Redundancy reflects the duplication of a system (i.e., the presence of isomorphic and isofunctional components) in order to ensure robustness in case of failure of the initial version (Mason, 2010). Conversely, the degenerate architecture of complex systems is isofunctional but heteromorphic (Mason, 2010; Tononi, Sporns, & Edelman, 1999), and supports both stability of the motor organization against perturbations, and functional variability in facing dynamical environment (Seifert et al., 2013). Specifically, neurobiological degeneracy exists at all levels of neurobiological systems (e.g., from genetic code to behavioural repertories) and is technically defined as 'the ability of elements that are structurally different to perform the same function or yield the same output' (Edelman & Gally, 2001, p. 13763). But unlike redundancy, degeneracy also provides the ability to perform

* Corresponding author at: Centre for Sports Engineering Research, Sheffield Hallam University, Collegiate Hall, Collegiate Crescent, Sheffield, S10 2BP, United Kingdom. Tel.: +33 688 893 451.
E-mail address: john.komar@me.com (J. Komar).
[1] CETAPS, Department of Sports Sciences, University of Rouen, France.
[2] Physical Education and Sports Sciences, National Institute of Education, Nanyang Technological University, Singapore.
[3] Centre for Sports Engineering Research, Sheffield Hallam University, United Kingdom.

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different functions under different conditions (Mason, 2010; Whitacre, 2010). Critically, degenerate elements may share the same function, but unlike redundant elements, they do not share the same over-all structure.

For instance, in a ski-simulator task, Hong and Newell (2006b) showed how different participants were able to use different inter-limb coordination patterns to reach the same performance outcomes. When knee motion relations were examined, both in-phase and anti-phase coordination patterns were exhibited by the participants with the presence of an effective coupling between the center of mass and the ski platform (i.e., the same effective outcome). From Hong and Newell (2006b), it was concluded that locally, different joint organizations could occur although similar global function was achieved. Similarly, Rein, Button, Davids, and Summers (2010) analyzed the full-body kinematics of basketball hook shoot and reported high inter-individual differences in elbow–shoulder and knee–hip coordination during the execution of the shots at the same target distance (with distances tested varying from 2 to 9 m). In addition, Rein, Button, Davids, and Summers (2010) also highlighted multiple pathways of adaptation with reference to different throwing distances. When performance of cricket batting was investigated, Pinder, Davids, and Renshaw (2012) demonstrated how individuals exploited system degeneracy to functionally adapt stable movement pattern solutions to satisfy changing task constraints (e.g., forward or backward movements). These results provided strong empirical support for the emergence of a functional degenerate behavior in multi-articular movements. Indeed, it appears that when several movement patterns are available, learners can individually exploit the inherent degeneracy present in neurobiological systems to achieve the task goal.

In addition to ensure stability against perturbations and adaptation to dynamical environment, it is suggested that the degenerate architecture of neurobiological systems plays a central role in adaptive innovation of complex system (Hristovski, Davids, & Araújo, 2006; Whitacre, 2010). It creates the potential for variations and encourages “pluripotentiality” (Mason, 2010, p. 281). Specifically, pluripotentiality refers to the potential involvement or removal of biomechanical degrees of freedom to increase stability or flexibility of the system. More precisely, there are opportunities for the human movement system to acquire two or more coordination patterns, sometimes partially sharing same elements (e.g., partially sharing same joints), to accomplish a task. For instance in catching a ball, one can essentially coordinate his shoulder, elbow and wrist to reach the ball, but could also move and coordinate his entire body while additionally involving his hip, knee and ankle to realize the task. This partial use of possible joints offers the possibility to always involve additional degrees of freedom into the overall coordinative structure to ensure task realization. In other words, degeneracy may allow a surplus of degrees of freedom necessary for later exaptation (Mason, 2010). Degeneracy therefore plays an important role in learning, providing the basis for the diversity of actions required to negotiate information-rich environment, as well as providing huge evolutionary fitness advantage (Seifert et al., 2013). Importantly, the acquired coordination is a product of self-organization and the instances of exploration (i.e., variability during learning) are an inherent part of the learning process (Newell, 1991; Sporns & Edelman, 1993). Chow, Davids, Button, and Koh (2008), in a soccer kicking study, further observed that there were actually no common pathways of change in coordination pattern during learning, suggesting that directional changes in degrees of freedom were dependent on task relevant intrinsic dynamics. Inevitably in the process of skill acquisition, the pathway to organize joints was different across participants, suggesting the functional role of degeneracy during learning.

One framework that could yield insights into the functional role of system degeneracy is Newell's model of learning with three stages of learning (*coordination*, *control* and *skill*, Newell, 1985). Early in learning (i.e. between coordination and control stage) novices are challenged to seek for a functional assembling of the body parts regarding task and environmental conditions. Once reached the control stage, performers are able to perform with consistency in changing performance environment. Later during learning, skill stage is observed when performers are able to use reactive forces from their muscular–skeletal system or from the environment to efficiently achieve the task. Specifically, for learners who are navigating between coordination and control stage of learning, there is a need to progressively become more attuned to the consequences of different combinations of key variables in impacting the task behaviour. Degeneracy may therefore support the explorative activity of learners involved in the transition between those two stages (see Chow et al., 2008).

Interestingly, swimming represents a supra-coordinative task, but it is assumed that a biomechanically effective pattern of coordination between elbow and knee oscillations seems to be adopted by expert performers compared to novices (Seifert et al., 2011). For example, experts were able to start a swimming cycle with arm–leg coordination in anti-phase followed by an in-phase mode and then back to anti-phase mode through every cycle of movement (i.e., every 1–2 s). In aquatic activities where the environmental constraints play an important role (e.g., due to high density of water), the use of a biomechanically effective pattern becomes pertinent for effective performances (Seifert, Leblanc, Chollet, & Delignières, 2010). In addition, the level of environmental constraint (i.e., the amount of forward resistance swimmers have to overcome in order to move) is related to the swimming speed squared (see Toussaint & Truijens, 2005), therefore an increase in swimming speed leads to quadratic increase of environmental constraints. It is therefore unclear whether learners need to actively explore the perceptual-motor workspace when learning a specific biomechanical expert pattern during their transition between coordination and control stage of learning, or instead converge directly towards this biomechanical expert pattern.

This paper aims to investigate the role of neurobiological degeneracy in the acquisition of coordination in a task where the perceptual-motor workspace is highly constrained and the existence of a biomechanically effective expert pattern is assumed. More precisely, the objectives were to: i) determine if there is inherent degeneracy for a swimming task; ii) determine how the level of constraints can limit the expression of degeneracy; and iii) explore how degeneracy can support a functional role for movement variability. We then hypothesized that even if the desired outcome of learning is the acquisition of a single biomechanically effective pattern, active exploration of the workspace can be facilitated by the inherent neurobiological degeneracy. In addition, it is predicted that participants who are seeking for a functional coordination (i.e., between coordination and control stage of learning, Newell, 1985) may exhibit different patterns of coordination reflecting explorative behaviours. Whereas stability and accuracy are usually the features of the movement to define an optimal learning (Tallet, Kostrubiec, & Zanone, 2008), the findings from this study may highlight the critical role played by neurobiological degeneracy and especially pluripotentiality property as a way of acquiring skill flexibility during the learning process.

2. Method

2.1. Participants

Fourteen breaststroke swimmers with an equivalent level of expertise and representing a range of recreational level swimmers (i.e., assessed by two experienced swimming instructors and motor control specialists as navigating between coordination and control stage of learning based on Newell's model of learning) were recruited for this study. Expertise level was expressed as a percentage of the current world record (W.R.) based on their best performance for 50-m breaststroke. Six women and eight men (18.9 ± 1.0 years, 62.2 ± 8.5 kg, 1.72 ± 0.63 m height; time measured on the day of the test for 25 m: 22.1 ± 2.5 s; best performance for 50 m: 38.89 ± 4.41 s that represents 63.70 ± 7.25%W.R.) were recruited for this study. The protocol, approved by the University ethics committee, was explained to the swimmers, who then gave their written informed consent to participate in this study.

2.2. Task and procedure

Participants were first required to warm up individually in a 25-m indoor pool at moderate intensity. Thereafter, the participants were asked to perform four trials, each consisting of a 25 m swim followed by 5 min rest between trials. The demands of the task were identical for each participant, i.e., normalized based on their personal maximal achievable speed. Two trials were performed in highly constrained environment (90% of their maximal speed) and two trials in weakly constrained environment (70% of their maximal speed). The speed of the trials was determined a priori and swimmers were then asked to self pace a target time during the 25 m swims. The order of speed conditions was randomized across participants. A margin of $\pm 5\%$ error on the time achieved compared to the target time was accepted as a valid trial, and participants had to repeat this trial if they were too slow or too fast, in order to establish two trials per speed condition that are acceptable for each individual.

2.3. Video recording

Black body markers were placed on the anatomical landmarks of the wrist (radiocarpial joint), elbow (ulnohumeral joint), shoulder (humeral head), hip (trochanter major of femur), knee (tibiofemoral joint) and ankle (talocrural joints) of the participants. A calibration frame of 4 m in horizontal-axis (X), 2 m in vertical axis (Y) and 2 m in lateral axis (Z) was positioned on the floor of the pool, in the middle of the pool and orthogonally to the wall. Two above-water and four underwater stationary cameras (Panasonic NV-GS17, 50 Hz) were positioned around the calibration frame according to a previous study by Psycharakis and Sanders (2009). The six views were synchronized and genlocked a posteriori with Adobe Premiere© 6.0. Cameras captured video clips of one cycle per swim taken in the central part of the pool. Fields of the cameras were crossed to allow views of all body markers with at least two cameras capturing any single marker at any time. Each cycle corresponded to the period from one maximal knee flexion to the next maximal knee flexion (Seifert et al., 2010).

2.4. Movement analysis

2.4.1. Angles calculation

Digitization of body markers on video data allowed 3-D reconstruction of body markers using APAS software (Ariel Dynamics) and allowed the calculation of relative elbow and knee angles (following Komar, Sanders, Chollet, & Seifert, 2014). Angular displacements were calculated as the arctangent of the dot product of the limb unit vector of two adjacent limbs with the standard corrections for quadrant. Angular velocities were then computed as the first derivative of the angular position using the central difference formula.

2.4.2. Computation of continuous relative phase

As previously introduced by Haken, Kelso, and Bunz (1985), the continuous relative phase (CRP) was considered as a suitable order parameter or collective variable representing the emergent organization of the multiple degrees of freedom (Seifert et al., 2010, 2011).

With reference to Hamill, Haddad, and Mcdermott (2000), the data on angular displacements and angular velocities were normalized at an interval $[-1, +1]$ as follows (see Eqs. (1) and (2)):

$$\text{Angular position}: \quad \theta_{\text{norm}} = \frac{2\theta}{(\theta\max - \theta\min)} - \frac{(\theta\max + \theta\min)}{(\theta\max - \theta\min)} \tag{1}$$

where θmax is the maximum angular position within one complete cycle and θmin is the minimum angular position within one complete cycle.

$$\text{Angular velocity}: \quad \omega_{\text{norm}} = \frac{2\omega}{(\omega\max - \omega\min)} - \frac{(\omega\max + \omega\min)}{(\omega\max - \omega\min)} \tag{2}$$

where ωmax is the maximum angular velocity within one complete cycle and ωmin is the minimum angular velocity within one complete cycle. Knee and elbow angles were previously filtered using a low-pass Fourier filter (cut off frequency at 6 Hz). Phase angles (ϕ_{elbow} and ϕ_{knee}) in degrees were subsequently calculated (see Eq. (3)) and corrected according to their quadrant (3):

$$\phi = \arctan(\omega_{\text{norm}}/\theta_{\text{norm}}). \tag{3}$$

Finally, CRP for a complete cycle was calculated as the difference between both phase angles (Eq. (4)):

$$\text{CRP} = \phi_{\text{elbow}} - \phi_{\text{knee}}. \tag{4}$$

Theoretically, two extreme modes of coordination are possible: in-phase (CRP $= 0°$) and anti-phase (CRP $= 180°$); however, following previous studies in inter-limb coordination, a lag of $\pm 30°$ was accepted to define the adopted coordination mode (e.g., Bardy, Oullier, Bootsma, & Stoffregen, 2002; Diedrich & Warren, 1998). Therefore, an in-phase mode was assumed to occur when $-30° < \text{CRP} < 30°$, while an anti-phase mode was taken to be between $-180° < \text{CRP} > -150°$ and $150° < \text{CRP} > 180°$. Following Seifert et al. (2011), the first CRP value of the cycle provides an indication about the capability of the swimmers to synchronize the start of the leg extension with regard to the end of the arm recovery. The time spent in in-phase corresponds to the total percentage of cycle duration spent in in-phase subtracted by the time spent with arms and legs outstretched. It indicates how the swimmers superposed iso-directional actions (i.e., leg flexion [corresponding to propulsion] with arm flexion [corresponding to recovery] or leg extension with arm extension). The maximal peak of CRP indicates the period when the legs lead the arms in flexion or extension (e.g., legs are early recovering when arms are extended). Finally, the mean value of CRP across the cycle indicates which oscillator globally leads the other (i.e., a positive value indicates that knee is leading and a negative value indicates that elbow is leading).

2.4.3. Estimation of movement outcome

The instantaneous position of the center of mass (CM) was based on the anatomical model adapted by De Leva (1996), with the consideration that the head is fixed relative to the trunk, the foot as fixed relative to the shank, and the hand as fixed relative to the forearm. In total, six anatomical points where digitized, but considering the symmetrical movements of the breaststroke pattern, these six points were assumed to accurately represent both sides of the individuals and therefore the most influential body parts on the CM in the X and Y axis. The intra-cycle velocity variations of the CM were then calculated as the first derivative of the displacement of the CM using the central difference formula. The intra-cyclic acceleration variations of the CM were calculated as the second derivative of the displacement of the CM using the central difference formula. For all the cycles, the mean swimming speed (v, in m s^{-1}) and the stroke rate (SR, in Hz) were calculated.

2.5. Dependent variables

2.5.1. Movement pattern and coordination variables

Five indicators of coordination were derived from the CRP in order to perform the cluster analysis (Seifert et al., 2011): the time spent in in-phase, the relative phase at beginning of the cycle, the maximal value of CRP, the standard deviation of the CRP across a cycle and the mean value of CRP across the cycle. In addition, variables about independent oscillators were used to characterize the different patterns: the elbow angle at the beginning of the cycle, the amplitude of elbow motion, the maximal value of elbow across the cycle, the amplitude of knee motion, the maximal value of knee across the cycle, the maximal knee rotation speed (i.e., during extension), the minimal knee rotation speed (i.e., during flexion), the maximal elbow rotation speed and the minimal elbow rotation speed.

2.5.2. Movement outcomes

Performance was defined by the mean swimming speed and stroke rate. In addition, four indicators of movement effects resulting from movement organization during specific phases of a cycle were investigated (see Leblanc, Seifert, Tourny-Chollet, & Chollet, 2007): the maximal acceleration of the CM during leg propulsion (i.e., extension), the maximal acceleration of the CM during arm propulsion (i.e., flexion), the distance covered by the CM during leg propulsion and the distance covered by the CM during arm extension.

2.6. Data analysis

2.6.1. Coordination profiling

Based on coordination variables, hierarchical agglomerative cluster analysis using the squared Euclidean distance dissimilarity measure and the Ward linkage method (Everitt, Landau, & Morven, 2001) was applied to determine several profiles within this group of swimmers. Before computation, a pre-processing procedure was required and all variables were normalized between the interval [0, 1] to avoid errors due to the different scales of the measures (Rein, Button, Davids, & Summers, 2010).

2.6.2. Cluster validation

According to Breiman (1996) and Rein, Davids, and Button (2010), the number of clusters and classification of the participants in the clusters were validated by two procedures, namely i) a bootstrapping procedure testing the stability of the clusters, and ii) the use of an index testing the compactness of clusters. One bootstrapping procedure was used to construct the dendrogram, and this was repeated for all participants − 1 (one excluded at the time) and determined if the obtained classifications are stable or not. For example, if participant 1 was removed from the analysis, it is critical to determine if each participant remained in his initial cluster or if there was a switch from one cluster to another cluster). The identical bootstrapping procedure was also applied to repeat the clustering with all coordination variables − 1. From the bootstrapping procedure, a classification was considered as optimal when the composition of the dendrogram (number of clusters and classification of the participants in the cluster) did not change in comparison to the initial result. In addition, as no real clustering was a priori defined, an internal estimator, the index of Calinski & Harabasz (Calinski & Harabasz, 1974), was used to estimate the number of clusters that best fitted the data (Zhao & Karypis, 2005). The index of Calinski Harabasz (CH) is defined by Eq. (5) and was calculated for a range from two to eight potential clusters:

$$\mathrm{CH}(k) = \frac{\mathrm{B}(k)/(k-1)}{\mathrm{W}(k)/(n-k)} \tag{5}$$

where n represents the number of participants, k denotes the number of clusters [2–8], and $\mathrm{B}(k)$ and $\mathrm{W}(k)$ denote the between and within clusters sums of squares of the partition. The maximal value of CH indicates optimal ratio between inter-cluster distance and intra-cluster distance and in turn, an optimal value of the number of clusters. All tests were conducted with MatLab r2010b.

2.6.3. Statistical analysis

For all measured variables, raw time-series of all cycles were time-normalized (100%) and therefore allowing comparison and averaging within clusters. After performing the cluster analysis, and when the normality of the distribution (Kolmogorov–Smirnov test) and the variance homogeneity (Levene's test) were confirmed to be acceptable, the comparison between emerging clusters was done by one-way ANOVA (fixed factor: group) and Tukey HSD post-hoc test was used in a case of more than two groups. Partial eta squared (η_P^2) was calculated as an indicator of effect size, considering that $\eta_P^2 = 0.02$ represents a small effect, $\eta_P^2 = 0.13$ represents a medium effect and $\eta_P^2 = 0.26$ represents a large effect (Cohen, 1988). All analyses were undertaken using SPSS (SPSS Statistics 20.0, IBM) with a conventional significance level at $p < .05$.

3. Results

3.1. Coordination profiling

In a weakly constrained environment (i.e., low speed condition), the cluster analysis enabled the classification of participants in four different clusters as shown in Fig. 1 (Cluster 1 = P7, P13, P12 and P14; Cluster 2 = P1, P8, P9 and P4; Cluster, 3 = P5, P6 and S11; Cluster 4 = P2, P3 and P10). The CH index was the highest for this number of cluster (CH = 3.2). The cluster validation using the bagging procedure showed that the cluster composition changed when participant 6 was removed (switch of participant 1 from cluster 3 to cluster 4), and when participant 8 and participant 13 were removed (switch of participant 9 from cluster 3 to cluster 4). Furthermore, the cluster composition did not change when the other participants were removed one by one.

For this weakly constrained condition, each cluster differed by their arm–leg coordination at the beginning of the cycle that is by the position of their arms more or less streamlined when they began propulsion of their legs. In addition, cluster 3 is characterized by high value of CRP corresponding to periods when legs are leading the arms, and a low standard deviation of coordination during the cycle. Cluster 2 is characterized by a high amount of time spent in in-phase coordination mode, corresponding to a superposition of a large amount of contradictory actions (i.e., arm propulsion and leg recovery or arm recovery and leg propulsion) and a low standard deviation of coordination during the cycle. Conversely, cluster 4 is characterized by a large standard deviation of coordination where individuals had the tendency to switch from anti-phase mode to in-phase mode to anti-phase mode within a cycle. Cluster 1 appears most similar to cluster 4 except for the CRP value at the beginning that was lower (−150° for cluster 4 and −118° for cluster 1).

More precisely, while looking at the differences in terms of coordination variables, each group appeared somehow significantly different from the others. The patterns of coordination of the four clusters presented in Fig. 2 (left panel) exhibited a different value of CRP at the beginning of the cycle ($F(3,10) = 6.54$, $p = .010$, $\eta_P^2 = .662$, all $p_s < .011$ for post-hoc). Therefore, each cluster showed a different level of arm extension when knees started their extension (i.e., leg propulsion). Namely, the closer to anti-phase the first CRP value (i.e., close to −180°) is, the more streamlined were the arms when the legs began propulsion. In addition, cluster 3 is characterized by more time spent with legs leading the arms, with the maximal value of CRP appearing higher for cluster 3 than the other clusters ($F(3,10) = 5.71$, $p = .015$, $\eta_P^2 = .632$, all $p_s < .026$ for post-hoc). Similarly, the mean value of CRP was positive for cluster 3 ($M = 23.9 \pm 3.3°$), when it was negative and significantly lower ($M = -17.8 \pm 7.8°$) for the other clusters than cluster 3 ($F(3,10) = 9.23$, $p = .003$, $\eta_P^2 = .735$, all $p_s < .006$ for post-hoc). This cluster 3 also showed a lower standard deviation of the CRP (37.5 ± 0.4°) than cluster 1 (61.3 ± 6.1°) and cluster 4 (74.6 ± 10.2°) ($F(3,10) = 11.56$, $p < .001$, $\eta_P^2 = .776$, all $p_s < .020$ for post-hoc). In addition, cluster 4 also differed from cluster 2 in terms of

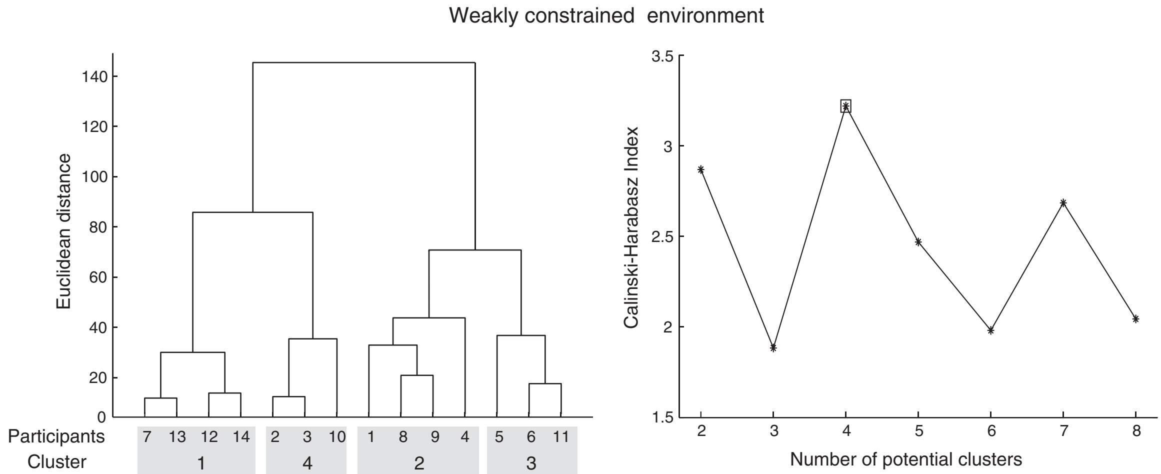

Fig. 1. Hierarchical agglomerative dendrogram (left panel) and index of Calinsky–Harabasz for 2 to 8 potential clusters (right panel) in the weakly constrained environment.

standard deviation ($p = 0.006$). Finally, cluster 1 differed from cluster 2 in terms of time spent in in-phase ($F(3,10) = 6.54$, $p = .010$, $\eta_P^2 = .662$, $p < .010$ for post-hoc).

In a highly constrained environment (i.e., high speed condition), the higher level of environmental constraint appeared limiting the expression of degeneracy as individuals expressed less possibility to vary their coordination. Indeed, only two different clusters in terms of the classification of performers were present from the cluster analysis undertaken and this is shown in Fig. 3 (Cluster 1 = P2, P4, P14, P10, P5, P7 and P12; Cluster 2 = P1, P8, P11, P3, P13, P6 and P9). The CH index was the highest for this number of cluster (CH = 10.5). The cluster validation using the bagging procedure showed that the cluster composition did not change when the performers were removed one by one. In this condition, Fig. 2 (right panel) showed that cluster 1 exhibited a lower value of CRP at the beginning of the cycle ($F(1,12) = 14.74$, $p = .002$, $\eta_P^2 = .575$) and a higher mean value of CRP than cluster 2 (mean CRP = $-37.1 \pm 11.1°$ for cluster 1 and $-59.4 \pm 12.2°$ for cluster 2) ($F(1,12) = 12.93$, $p = .004$, $\eta_P^2 = .519$). For all the other variables (i.e., time spent in in-phase, maximal value of CRP and standard deviation of CRP), no differences between clusters were found.

3.2. Movement patterns

In addition to coordination profiling, Fig. 4 and Table 1 present the characteristics of each oscillator. In the weakly constrained condition, the differences between clusters appeared mainly on the elbow rather than on the knee. Indeed, individuals exhibited a similar knee pattern in both weakly and highly constrained conditions. The only exception is the presence of a higher knee angular velocity during extension (i.e., propulsion) for clusters 3 and 4 in a weakly constrained environment ($F(3,10) = 18.81$, $p = .001$, $\eta_P^2 = .849$, $p_s < .048$ for post hoc) (Table 1).

Conversely, clusters are differentiated by the movement pattern of the elbow that appeared as a key factor of the coordination. More precisely, in weakly constrained environment, cluster 4 exhibited higher elbow angular velocity during extension (i.e., recovery) ($F(3,10) = 4.78$, $p = .026$, $\eta_P^2 = .590$, $p_s < .048$ for post hoc), but lower amplitude of elbow movement ($F(3,10) = 5.08$, $p = .022$, $\eta_P^2 = .603$, $p = .016$ for post hoc) as observed from the lowest maximal value reached during complete extension ($F(3,10) = 3.96$, $p = .042$, $\eta_P^2 = .543$, $p = .031$ for post hoc). In addition, different values of elbow angle at beginning of the cycle were revealed ($F(3,10) = 9.51$, $p = .003$, $\eta_P^2 = .741$), validating the previous results observed from the CRP value at the beginning of the cycle. Namely, cluster 3 showed a lower value of elbow angle at the beginning of the cycle ($p_s < .019$) while cluster 4 showed higher value ($p_s < .033$), which resulted in more streamlined arms during the beginning of the cycle (i.e. during leg extension). Finally, cluster 1 exhibited a higher elbow angular velocity ($F(3,10) = 4.45$, $p = .031$, $\eta_P^2 = .572$) compared to cluster 2 ($p = 0.027$).

As previously highlighted in coordination profiling, when performing in highly constrained environment, individuals exhibited less possibilities to vary their behaviour. In this condition, individuals with cluster 1 movement pattern primarily exhibited a lower value of elbow angle at the

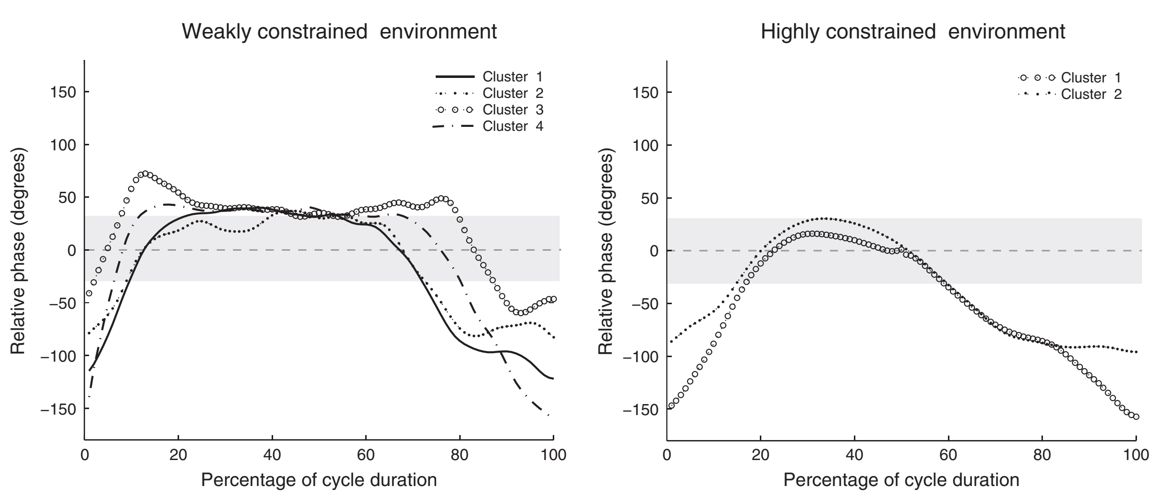

Fig. 2. Mean patterns of CRP for both weakly (left) and highly (right) constrained environment (the gray zone represents the in-phase coordination mode).

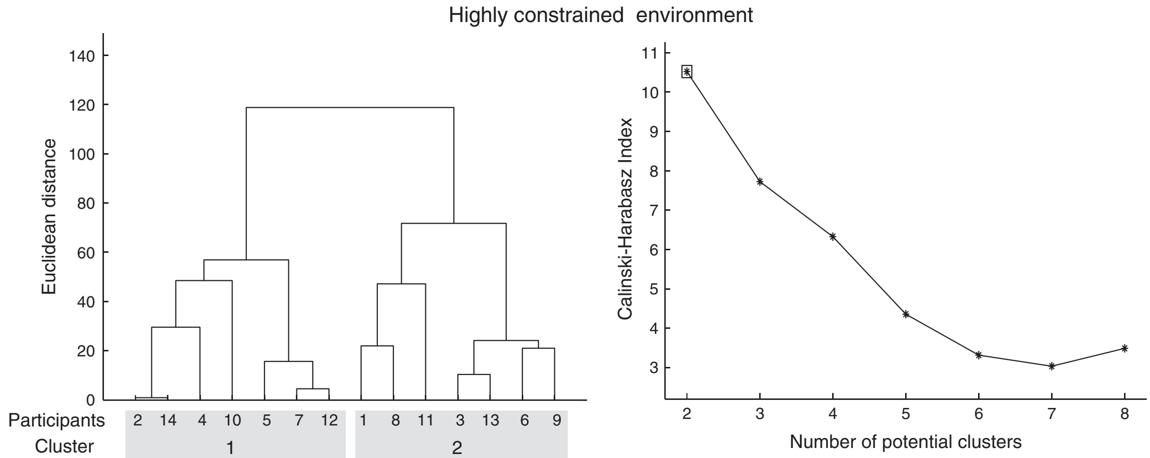

Fig. 3. Hierarchical agglomerative dendrogram (left panel) and index of Calinsky–Harabasz for 2 to 8 potential clusters (right panel) in the highly constrained environment.

beginning of the cycle ($F(1,12) = 29.48$, $p < .001$, $\eta_P^2 = .711$), which could be accounted for by the lower value of elbow angular velocity during extension ($F(1,12) = 12.77$, $p = .004$, $\eta_P^2 = .516$). Indeed, this lower value of angular velocity during extension could be the reason for the lag in reaching the maximal elbow extension.

3.3. Performance outcomes

The study of performance outcomes resulting from the adopted coordination pattern revealed that in both weakly and highly constrained environments, the different clusters of movement demonstrated the

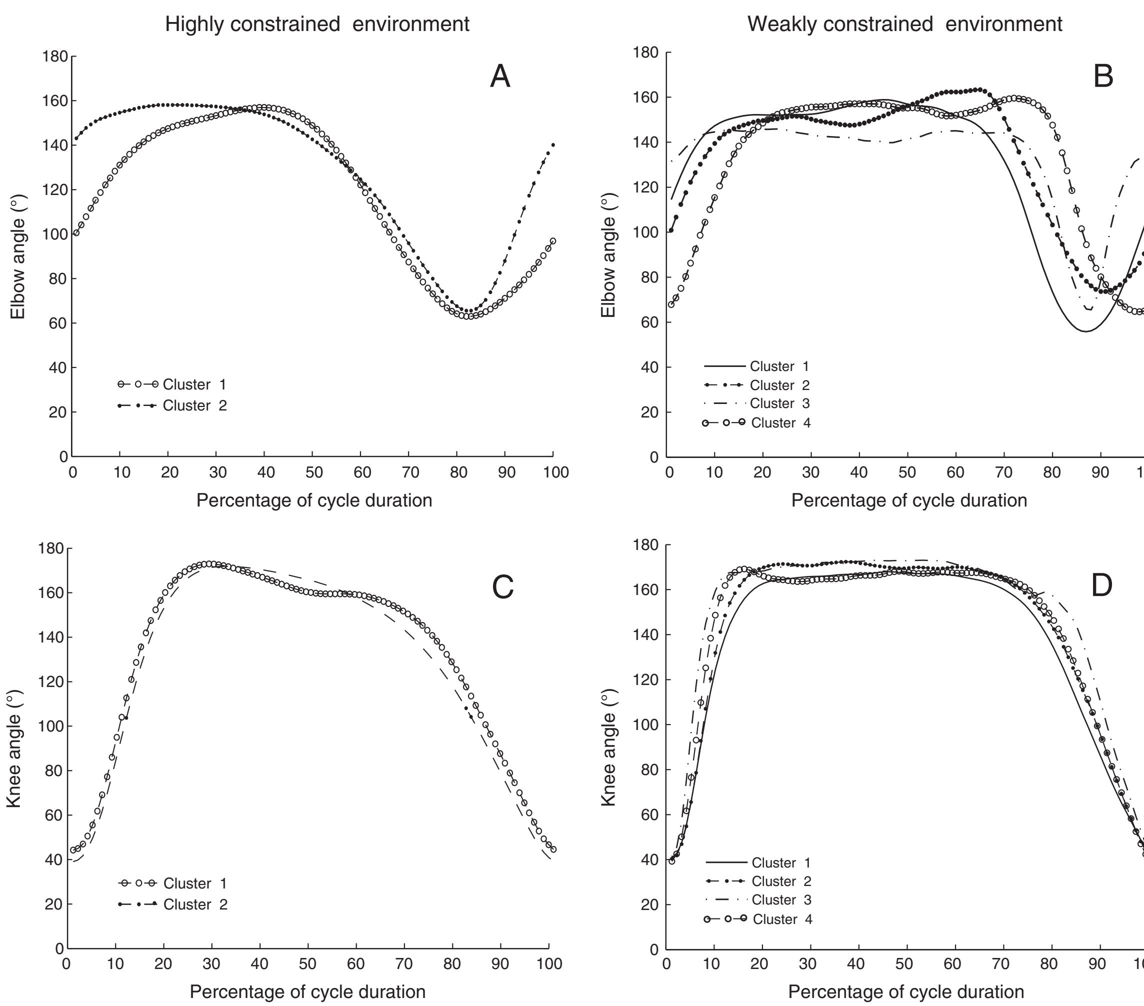

Fig. 4. Mean patterns of elbow angular positions in highly constrained environment (A) and in weakly constrained environment (B), and mean patterns of knee angular positions in highly constrained environment (C) and in weakly constrained environment (D).

Table 1
Mean and *SD* of movement patterns variables as a function of cluster and level of environmental constraints.

Outcome variables	Highly constrained environment				Weakly constrained environment							
	Cluster 1		Cluster 2		Cluster 1		Cluster 2		Cluster 3		Cluster 4	
	M	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Elbow at beginning (°)	100.57	13.42 *	143.06	15.78	114.46	19.23	101.48	11.41	67.84[a]	8.22	136.83[b,c]	14.80
Elbow angle amplitude (°)	103.05	13.27	107.56	10.42	113.76	6.96	96.33	12.85	97.46	6.70	85.86[a]	6.27
Maximal elbow angle (°)	160.93	6.14	163.96	6.45	164.17	7.44	167.22	7.57	161.06	3.84	150.52[b]	5.20
Maximal knee angle (°)	171.24	2.97	172.25	4.55	170.11	4.29	173.22	3.10	168.88	6.64	173.69	2.05
Knee angle amplitude (°)	133.16	5.23	129.40	5.01	127.15	4.90	130.89	6.11	127.70	11.72	130.38	8.24
Knee angular velocity during flexion ($° s^{-1}$)	−331.75	72.72	−373.79	44.41	−212.93	28.89	−222.54	41.55	−219.49	30.14	−220.63	29.77
Knee angular velocity during extension ($° s^{-1}$)	701.63	58.43	719.47	89.27	503.12	40.87	494.85	36.74	601.05[a,b]	3.44	674.45[a,b]	38.60
Elbow angular velocity during extension ($° s^{-1}$)	342.45	53.10*	505.67	108.52	257.70	85.27	227.65	59.44	210.99	26.77	376.39[b,c]	86.32
Elbow angular velocity during flexion ($° s^{-1}$)	−370.95	75.01	−408.70	35.59	−367.70	33.03	−277.71[a]	47.02	−292.70	24.90	−293.33	32.91

Note. In weakly constrained environment: a = significantly different from Cluster 1; b = significantly different from Cluster 2; c = significantly different from Cluster 3. In highly constrained environment: * = significantly different from Cluster 2; for all $p < 0.05$.

same mean swimming speed (Table 2). Similarly, no differences were revealed in terms of stroke rate.

3.4. Movement effect

Finally, despite the clusters showing different coordination patterns in weakly constrained environment, no significant difference was found between clusters in terms of movement effects, except for cluster 4 that showed a higher peak for acceleration of the CM during arm propulsion as compared to clusters 1 and 2 ($F(3,10) = 5.11$, $p = .021$, $\eta_P^2 = .0.605$, all $p_s < .031$ for post-hoc) (Table 2). It should be noted that this higher peak of acceleration did not lead to higher distance covered during arm propulsion. For all the variables relating to the leg propulsion phase (i.e., acceleration peak of the CM and distance covered by the CM), no differences were found. In highly constrained environment, no differences were found for movement effects. Indeed, for both acceleration of the CM during leg propulsion and arm propulsion, and for the distance covered by the CM during these two phases, cluster 1 and cluster 2 exhibited no significant differences.

4. Discussion

The aim of this study was to examine degeneracy in coordination modes in relation to movement effects (i.e., acceleration peak and distances covered by CM during propulsive phases) and performance outcomes (i.e., swimming speed and stroke rate) in a task with an emphasis on movement form. Instead of traditional cohort studies, which focus on expert–novices comparisons, we investigated coordination variability for participants who are actually in the process of acquiring a functional coordination (i.e., between coordination and control stage of learning).

4.1. System's degeneracy regarding task realization, performance and movement effects

Based on previous work in both discrete and cyclic multi-articular actions (e.g., Hong & Newell, 2006a; Pinder et al., 2012; Rein, Davids, & Button, 2010), the present findings revealed evidence of system degeneracy (Edelman & Gally, 2001), with cluster analysis performed on coordination data highlighting the existence of different motor organizations for an identical task. In both highly and weakly constrained environment, participants showed two and four different patterns of coordination respectively. In addition to previous studies where the outcome in the task was binary, namely "realized" or "not realized" (e.g. Hong & Newell, 2006a), we tried to go further in the definition of "same function or same output" (Edelman & Gally, 2001, p. 13763) by measuring task performance. In this idea, neurobiological degeneracy was obvious as no difference in terms of performance in the task was revealed between groups (e.g. speed of task realization). In the same idea, results showed that different patterns led to the same movement effects (i.e., acceleration and displacement of the CM). Only an exception occurred under the weakly constrained environment where cluster 4 exhibited a higher peak of acceleration of the CM during arm flexion (i.e., arm propulsion), but the distance covered by the CM captured within this cluster during arm flexion did not appear significantly higher. It should also be noted that cluster 4 exhibited the coordination pattern closest to that seen in the expert pattern (Seifert et al., 2011). However, the absence of higher performance outcome observed in participants demonstrating this pattern is indicative that they have not yet moved beyond the control stage of learning. Overall, results highlighted that individuals can use different coordination not only while performing the same task (i.e. breaststroke swimming), but also while performing this task at the same speed and with identical movement effects.

Table 2
Mean and *SD* of movement outcomes variables as a function of cluster and level of environmental constraints.

Variables	Highly constrained environment				Weakly constrained environment							
	Cluster 1		Cluster 2		Cluster 1		Cluster 2		Cluster 3		Cluster 4	
	M	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Stroke rate (Hz)	0.72	0.11	0.73	0.15	0.36	0.06	0.34	0.06	0.33	0.02	0.29	0.04
Swimming speed (m)	0.96	0.08	1.02	0.11	0.62	0.06	0.59	0.04	0.64	0.09	0.68	0.08
Maximum acceleration during leg propulsion ($m s^{-2}$)	2.54	0.71	2.18	0.71	1.13	0.48	1.19	0.22	1.59	0.26	1.55	0.41
Maximum acceleration during arm propulsion ($m s^{-2}$)	1.39	1.01	1.41	0.78	0.40	0.26	0.41	0.21	0.71	0.22	0.98[a,b]	0.23
Distance covered during leg propulsion (m)	0.44	0.12	0.48	0.13	0.43	0.07	0.40	0.03	0.45	0.12	0.40	0.07
Distance covered during arm propulsion (m)	0.27	0.20	0.33	0.17	0.23	0.06	0.26	0.08	0.47	0.12	0.51	0.14

Note. In weakly constrained environment: a = significantly different from Cluster 1; b = significantly different from Cluster 2; for all $p < 0.05$.

4.2. System's degeneracy supporting motor exploration

Even in highly constraining activities like swimming, where the use of a specific effective pattern is assumed to be critical in achieving performance expertise, the neurobiological system's degeneracy may be highly functional in offering the flexibility to develop the required individualized coordination patterns (Seifert et al., 2013). Therefore, for learners seeking to develop a functional coordination, system's degeneracy highlighted in this experiment could support the functional role of movement variability while allowing motor exploration during learning. The increase in motor variability early during learning has been highlighted by Vereijken, van Emmerik, Whiting, and Newell (1992) on learning a ski simulator task. Specifically, these authors showed that learners go through a preliminary phase of freeing the degrees of freedom so that the to-be-learned coordination can settle and stabilize (from coordination to control stage), thereby allowing an improved performance in increasing the amplitude of movement on the ski platform (and therefore eventually moving beyond the control stage of learning to the skill stage). Vereijken et al. (1992) therefore showed a non-linear relationship between coordination and performance outcome that may highlight the critical role of degeneracy during learning. Indeed, in the present study, even when participants navigating between coordination and control stage of learning exhibited the same performance outcome, the different profiles of coordination were associated with different higher order derivatives supporting performance. Particularly, the different emerging clusters differed in knee angular velocity during extension (i.e., propulsion) and elbow angular velocity during extension and flexion (i.e., propulsion and recovery). Therefore, the different patterns of coordination exhibited by swimmers may be due to the variation in *how they explore movement patterns* by manipulating higher order derivatives like angular velocities or angular amplitude. As highlighted by Chow, Davids, and Button (2007) in a soccer kicking study, novice participants did not show the ability to vary higher order derivatives like foot velocity at ball contact as compared to skilled participants. Clearly, neurobiological degeneracy plays a paramount role in defining *explorative learning*, by allowing performers (especially between coordination and control stage of learning) to explore and discover functional relationships between manipulating higher order derivatives and meeting the specific task demands.

Interestingly at the level of individual oscillators, results showed that only one oscillator primarily offered the potential for exploration. Indeed, results showed that differences in inter-limb coordination were mostly due to differences in elbow patterns, rather than in knee patterns. This result is in line with previous results on four-limb coordination from Kelso and Jeka (1992), which reported how the arms are most likely to influence the overall coordination in four-limb movements. Kelso and Jeka (1992) highlighted that the transition from anti-phase to in-phase mode resides mainly in a "down" transition, namely a transition exhibited by the arms gaining a half cycle of movement to the leg (i.e., instead of an "up" transition in which leg moves faster than the arm for one or a few cycles in order to gain a cycle) (Kelso & Jeka, 1992). This phenomenon principally occurred at oscillation frequency higher than 1.75 Hz in Kelso's work, which is higher than the present frequency of elbow and knee oscillations (i.e., around 0.35 and 0.75 Hz).

4.3. System's degeneracy impacted by the level of interacting constraints

Interestingly, this difference in oscillation frequency may reflect the constraining nature of the environment in breaststroke swimming, and presents a strong case to highlight the role played by the interacting constraints acting on participant's behaviour in this swimming task (i.e., interaction of task, environmental and organismic constraints, Newell, 1986). Particularly, the high level of environmental constraint in aquatic locomotion may interact with the task constraint (i.e., frequency of oscillation) which can account for the increase predominance of "down" transitions. More generally when the different swimming conditions were compared (i.e., highly and weakly constrained environment), the number of emerging clusters appeared lower when the constraints were higher (two clusters in highly constrained environment and four clusters in weakly constrained environment respectively). One of the main explanations of this result might be the higher level of forward resistance a swimmer has to face when the speed is increased. Indeed, even if both experimental conditions could be defined as constraining (i.e. aquatic environment, imposed swimming style, imposed speed), Kolmogorov, Rumyantseva, Gordon, and Cappaert (1997) have shown specifically in breaststroke that forward resistance increases a lot with swimming speed (i.e. goes from 20 N when speed is 0.9 m s^{-1} to 90 N when the speed is 1.5 m s^{-1}).

This result is congruent with the idea of interacting constraints that channel emergent coordination, suggesting that the potential variations of coordination pattern can be altered by the level of constraints acting on performers (Newell, 1986). In paradigmatic works on bimanual coordination (see Kelso, 1995), it has already been shown that the level of task constraint (frequency of finger-oscillations) can limit the possible number of emerging coordination pattern (i.e., from multi-stability to mono-stability). As suggested by Hong and Newell (2006b), a large number of degrees of freedom offer the potential for degeneracy to surface. However, the present study also showed that the variation in the level of interacting constraints also plays an important role in offering the potential for degeneracy.

4.4. System's degeneracy supporting the potential for motor re-organization

The results from the present study highlighted the pluripotentiality of dynamical degrees of freedom that degeneracy offers to neurobiological systems. Indeed, results showed that cluster one in a weakly constrained environment significantly differs to the other clusters in terms of amplitude of elbow angle, validating that altering the involvement of mechanical degrees of freedom is one mechanism for degeneracy to surface (Hong & Newell, 2006b; Vereijken et al., 1992). In addition, the absence of different involvement of mechanical degrees of freedom for the other clusters in both highly and weakly constrained environment suggests that interacting constraints affect not only these degrees of freedom, but also the spatiotemporal organization of these mechanical degrees of freedom (Hong & Newell, 2006a).

The present study also showed that a possible channel for degeneracy to emerge resides in the pathway in which the same elements (i.e., biomechanical degrees of freedom) of neurobiological systems can be organized in different ways. This observation is reflected by the non-linear correspondence between clusters in weakly constrained environment to highly constrained environment. More precisely, the use of one pattern in weakly constrained environment did not necessarily define the nature of the pattern used in highly constrained environment and vice versa. Indeed, as shown in Fig. 5, each cluster in one

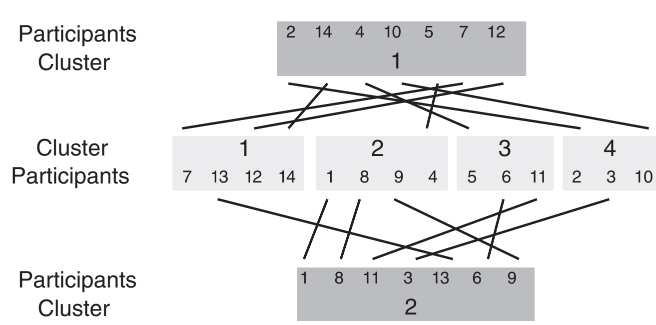

Fig. 5. Correspondence between clusters' participants in highly constrained environment (dark gray) and weakly constrained environment (gray).

environmental condition includes participants from each of the clusters present in the other environmental condition. Therefore, in a given situation, diverse sets of components can be recruited and organized to achieve the same function (i.e., many-to-one mapping) (Whitacre, 2010), but a set of components can also be recruited and organized differently in order to functionally respond to interacting constraints (i.e., one-to-many mapping). In other words, pluripotentiality observed here offers two sources of variability that appears highly intertwined, namely a source of variability for exploration during learning and a source of variability for adaptation to changing environment.

By offering the potential for the involvement of additional mechanical degrees of freedom, as well as the potential for non-linear reorganization of dynamical degrees of freedom, degeneracy therefore offers opportunities for later exaptations and learning.

5. Conclusion

The previous discussed results support the following conclusion: i) even in a highly constrained activity where an assumed specific pattern is required to acquire expertise, degeneracy is omni-present as evident by the different patterns of coordination exhibited by learners who are seeking to develop a functional movement pattern (i.e., between coordination and control stage of learning). However, the effect of the level of environmental constraint on the expression of degeneracy questions the existence of degeneracy at the boundaries of human performance, namely that degeneracy might disappear at maximal swimming speed; ii) this degenerate behavior seemed mostly reflected by variability of one oscillator, namely the elbow which seems to direct the non-homologous limb coordination; iii) emerging clusters differed in terms of higher order derivatives like elbow and knee rotation speed, and the degenerate behavior may help to define *explorative learning*, as the exploration of the relationship between these higher order derivatives and interacting constraints; iv) degeneracy allows both the stability of a function through a many-to-one mapping, but also a potential novelty through a one-to-many mapping such that a set of components can be reorganized differently (with possible additional involvement of degrees of freedom). Therefore, in motor skill acquisition, degeneracy can have a functional role in both ensuring stability as well as flexibility of a to-be-learned coordination pattern so that learners can be encouraged to explore the key relations between higher order derivatives and interacting constraints.

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